Humanitarian Logistics Hub Network Design Under Uncertainty: A Multi-Objective Model and Priority-Based Genetic Algorithm for Central Coastal Vietnam

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Abstract. Conventional disaster relief models often assume uninterrupted road networks, making them inadequate for Central Vietnam, where frequent floods severely disrupt infrastructure. We formulate the Multi-Objective Incomplete Hub Location Network Design Problem (MO-IHLNDP), a two-stage stochastic program that jointly minimizes expected logistics cost and expected maximum deprivation cost over a set of flood scenarios. The model explicitly accounts for broken inter-hub links, multi-modal transport (land, water, and air), pre-positioned inventory, as well as adaptive lateral transshipments and reactive hub activations to recover from severe localized deficits. To solve MO-IHLNDP, we propose PB-NSGA (Priority-Based Nondominated Sorting Genetic Algorithm), which confines the chromosome to first-stage hub decisions and reconstructs all scenario-dependent second-stage variables through a shared priority-based decoder, reducing search-space complexity from $O(|\mathcal{S}| \times |\mathcal{V}|)$ to $O(|\mathcal{H}| + |\mathcal{I}|)$. Experiments on standard benchmarks and a realistic CV-Small instance validate near-optimal accuracy against exact Pareto fronts. On the full-scale CV-Large case study under matched 20-seed run budgets, PB-NSGA uniquely achieves 100% out-of-sample feasibility on adversarial extreme scenarios and delivers a quantified cost-deprivation trade-off and a stable hub investment hierarchy for four flood-prone provinces in Central Vietnam.

Keywords: Humanitarian logistics · Hub location · Multi-objective optimization · Stochastic programming · Disaster relief · Evolutionary algorithm

1 Introduction

1.1 Problem

Natural disasters are becoming increasingly frequent and violent, continuously exposing supply chain weaknesses and causing massive economic losses [1].

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As a highly vulnerable coastal country, Vietnam has witnessed unprecedented extreme weather events. In 2025 alone, a relentless sequence of typhoons and record-breaking floods in Central Vietnam resulted in massive fatalities and an estimated 98.7 trillion VND in economic losses [2].

Humanitarian operations during this period faced significant challenges. Most critically, national logistics infrastructure suffered severe disruption: the North-South Railway was paralyzed by track erosion and submersion, forcing the cancellation of 170 trains, while key arteries such as Vietnam National Route 1 were severed by landslides, isolating communities for days [2]. These events demonstrate that during a major catastrophe, the assumption of a fully connected transport network is fundamentally flawed – the physical network becomes "incomplete", characterized by broken links and isolated nodes. Compounding this, resource scarcity arising from poor planning and uncertain supply and demand further strains relief efforts. Disaster Relief Network Design (DRND) bridges the preparedness and response phases by optimizing facility locations and relief routing [3].

1.2 Related Works

Humanitarian logistics prioritises speed and equity over cost minimisation [4]. Under uncertainty, stochastic programming and robust optimisation are the dominant paradigms [5,6], with deprivation cost—an exponential penalty on victim waiting time—capturing social welfare [7]. Hub-and-spoke structures consolidate relief flows efficiently [8]; for disaster settings, *incomplete* topologies in which infrastructure damage severs inter-hub arcs are more realistic [9]. Sangsawang and Chanta [10] established the flood-specific hub-location baseline (single-objective, deterministic, Thailand), and multi-modal extensions further address heterogeneous transport modes [11].

Two-stage stochastic programming separates pre-disaster design from post-disaster recourse [12]; robust formulations and prepositioning models outperform deterministic baselines under demand and supply uncertainty [13, 14]. NSGA-II [15] is the standard multi-objective framework in humanitarian logistics, valued for parameter-free Pareto ranking and elitist selection [11, 16].

Research gap and contributions. To the best of our knowledge, no existing model simultaneously addresses all four of the following: (1) incomplete hub networks with severed inter-hub arcs, (2) two-stage stochastic programming under flood uncertainty, (3) bi-objective optimisation of logistics cost *and* deprivation cost, and (4) a calibrated flood-disaster case study for Central Vietnam. Deterministic models [10] are the closest precedent: they assume full network connectivity and optimise a single cost objective, causing them to fail structurally when floods sever road links—the very scenario we target. Stochastic models [12, 13] capture demand and supply uncertainty but do not model link failures in the hub sub-network, and omit the deprivation cost that captures social equity across victim groups. This paper closes all four gaps simultaneously by formulating MO-IHLNDP and solving it via PB-NSGA, a priority-based [17] evolutionary algorithm whose encoder-decoder design efficiently handles the coupled first- and second-stage decisions of the two-stage stochastic program.

2 Problem Description and Mathematical Model

We model the network design problem as a capacitated, single-allocation MO-IHLNDP. The network comprises **origins** (supply sources), **demands** (victim groups), and **hubs** of two types: planned (pre-disaster) and reactive (post-disaster, dynamically opened).

From a practical standpoint, DRND involves two critical uncertainty factors. First, infrastructure disruption – damaged roads and buildings – can disable pre-programmed plans, necessitating a dynamic solution [13]. Second, the exact locations of demands, the total number of victims, and the availability of origins are all highly uncertain during rescue operations [12]. To account for these, we employ a two-stage stochastic programming approach across multiple scenarios, each assigning a specific risk factor per area, and assume that Search and Rescue operations rely heavily on dynamic external supply sources, as pre-positioned hubs may not fully satisfy uncertain demands under extreme scenarios.

The model integrates the deprivation cost [7] for social equity, and Daganzo's continuous approximation (CA) [18] to tractably estimate last-mile rescue times.

2.1 Notation

Sets: $\mathcal{H}$, $\mathcal{I}$, $\mathcal{J}$ index candidate hubs ($k, h \in \mathcal{H}$), demand nodes ($i \in \mathcal{I}$), and relief origins ($j \in \mathcal{J}$); $\mathcal{S}$ and $\mathcal{M}$ index disaster scenarios and transport modes; $\mathcal{V} = \mathcal{H} \cup \mathcal{I} \cup \mathcal{J}$ is the full node set.

First-phase parameters (pre-disaster, strategic): fixed hub establishment cost F_k; unit holding cost c_k and inventory capacity κ_k; inter-hub flow discount α; risk acceptability threshold χ; arc transport cost and travel time C_{uvm}, τ_{uvm}; local routing cost C_m, vehicle capacity Q_m per mode, and relief-items-per-person conversion ratio γ.

Second-phase parameters (post-disaster, scenario s with probability π_s): reactive hub activation cost F^a_{ks}; hub round-trip processing time τ_{ks}; arc accessibility $a_{uvms} \in \{0, 1\}$ (1 if traversable under scenario s); node risk index r_{us}; deprivation sensitivity $\lambda_{is} = \lambda_0(1 + r_{is})$, up-weighting vulnerable populations; demand D_{is} and supply O_{js}; pre-computed cheapest transport cost $c_{jks} = \min_{m|a_{jkms}=1} C_{jkm}$ and Daganzo CA last-mile routing cost Θ_{kis} (circuity Φ, group size η, grid area A_i).

Decision variables: $x_k \in \{0, 1\}$ and $y_{ks} \in \{0, 1\}$ activate planned and reactive hubs; $z_{iks}, z_{jks} \in \{0, 1\}$ assign demand nodes and origins to hubs per scenario; $q_k \geq 0$ is pre-positioned inventory quantity; $f_{khms} \geq 0$ is inter-hub lateral transshipment flow.

2.2 Objectives

Objective 1: Minimize Total Expected Logistics Cost (Z_1)

$$\begin{aligned}
\text{Min } Z_1 = & \underbrace{\sum_{k\in\mathcal{H}} F_k x_k + \sum_{k\in\mathcal{H}} c_k q_k}_{\text{Pre-disaster strategic costs}} \\
& + \sum_{s\in\mathcal{S}} \pi_s \Bigg[\underbrace{\sum_{k\in\mathcal{H}} F^a_{ks} y_{ks}}_{\text{Reactive setup}} + \underbrace{\sum_{j\in\mathcal{J}} \sum_{k\in\mathcal{H}} c_{jks} \cdot O_{js} z_{jks}}_{\text{Origin to Hub supply flow}} \\
& + \underbrace{\sum_{k\in\mathcal{H}} \sum_{h\in\mathcal{H}} \sum_{m\in\mathcal{M}} \alpha C_{khm} f_{khms}}_{\text{Inter-hub lateral trans-shipment}} + \underbrace{\sum_{k\in\mathcal{H}} \sum_{i\in\mathcal{I}} \Theta_{kis} z_{iks}}_{\text{Grid-based Last-Mile CA}} \Bigg]
\end{aligned} \tag{1}$$

where Θ_{kis} is the pre-computed Daganzo CA routing cost [18] (line-haul plus local detour):

$$\Theta_{kis} = \min_{m\in\mathcal{M}|a_{ikms}=1} \left(2 \cdot C_{kim} \cdot \lceil D_{is}/Q_m \rceil + C_m \cdot \Phi \sqrt{\lceil D_{is}/\eta \rceil \cdot A_i} \right)$$

Objective 2: Minimize Expected Maximum Deprivation Cost (Z_2)

$$\text{Min } Z_2 = \sum_{s\in\mathcal{S}} \pi_s \left(\max_{i\in\mathcal{I}} \left[D_{is} \cdot \left(e^{\lambda_{is}\cdot\Omega_{is}} - 1 \right) \right] \right) \tag{2}$$

$$\Omega_{is} = \sum_{k\in\mathcal{H}} z_{iks} \left(\tau_{ks} + 2 \cdot \min_{m\in\mathcal{M}|a_{ikms}=1} \tau_{kim} \right)$$

The exponential penalty [7] on waiting time Ω_{is} enforces equity; $\lambda_{is} = \lambda_0(1+r_{is})$ up-weights vulnerable populations. Exploiting single-allocation, C^{dep}_{iks} is pre-computed and Z_2 linearized via $W_s \geq C^{\text{dep}}_{iks} \cdot z_{iks}$:

$$C^{\text{dep}}_{iks} = D_{is} \cdot \left(e^{\lambda_{is}\cdot\left(\tau_{ks} + 2\cdot\min_{m\in\mathcal{M}|a_{ikms}=1} \tau_{kim}\right)} - 1 \right) \tag{3}$$

The auxiliary variable $W_s \geq C^{\text{dep}}_{iks} \cdot z_{iks}$ linearizes the max; the complete model minimizes Z_1, Z_2 subject to:

2.3 Constraints

$$x_k + y_{ks} \leq 1 \quad \forall k \in \mathcal{H}, s \in \mathcal{S} \tag{4}$$

$$q_k \leq \kappa_k \cdot x_k \quad \forall k \in \mathcal{H} \tag{5}$$

$$r_{ks} \cdot (x_k + y_{ks}) \leq \chi \quad \forall k \in \mathcal{H}, s \in \mathcal{S} \tag{6}$$

$$\sum_{k\in\mathcal{H}} z_{iks} = 1,\ \sum_{k\in\mathcal{H}} z_{jks} = 1 \quad \forall i\in\mathcal{I},\ j\in\mathcal{J},\ s\in\mathcal{S} \tag{7}$$

$$z_{iks},\ z_{jks} \leq x_k + y_{ks} \quad \forall i, j, k, s \tag{8}$$

$$z_{iks} \leq \textstyle\sum_m a_{ikms},\ z_{jks} \leq \sum_m a_{jkms} \quad \forall i, j, k, s \tag{9}$$

$$f_{khms} \leq M \cdot a_{khms} \quad \forall k, h \in \mathcal{H},\ m \in \mathcal{M},\ s \in \mathcal{S} \tag{10}$$

$$\begin{aligned}\sum_{i\in\mathcal{I}} \gamma D_{is} z_{iks} + \sum_{h\in\mathcal{H}} \sum_{m\in\mathcal{M}} f_{khms} \leq q_k + \sum_{j\in\mathcal{J}} O_{js} z_{jks} \\ + \sum_{h\in\mathcal{H}} \sum_{m\in\mathcal{M}} f_{hkms} \quad \forall k \in \mathcal{H}, s \in \mathcal{S}\end{aligned} \tag{11}$$

$$q_k + \sum_{j\in\mathcal{J}} O_{js} z_{jks} + \sum_{h\in\mathcal{H}} \sum_{m\in\mathcal{M}} f_{hkms} \leq \kappa_k \cdot (x_k + y_{ks}) \quad \forall k \in \mathcal{H}, s \in \mathcal{S} \tag{12}$$

$$x_k, y_{ks}, z_{iks}, z_{jks} \in \{0, 1\} \quad \forall i, j, k, s \tag{13}$$

$$q_k, f_{khms}, W_s \geq 0 \quad \forall k, h, m, s \tag{14}$$

Constraints (4)–(6) govern hub establishment and safe-zone placement; (7)–(9) enforce single-allocation via available arcs; (10)–(12) maintain flow feasibility and capacity; (13)–(14) define variable domains.

3 Proposed Algorithm: PB-NSGA

The MO-IHLNDP is NP-hard, as its single-objective, single-scenario relaxation subsumes the uncapacitated facility location problem [19]. We therefore propose **PB-NSGA** (Priority-Based Nondominated Sorting Genetic Algorithm), which embeds the NSGA-II framework [15] with a custom heuristic decoder. The decoder is algorithm-agnostic: the same reconstruction and evaluation core is also embedded in GWO-HD and VNS-TS, so cross-algorithm differences later reflect search behavior rather than different recourse logic. We embed NSGA-II [15] rather than MOEA/D [20]: MOEA/D requires pre-specified decomposition weight vectors, presupposing the Pareto front geometry a priori; NSGA-II's crowding-distance selection preserves spread across the full Z_1–Z_2 spectrum without weight tuning; and the encoder-decoder design is algorithm-agnostic—the *decoder* is the contribution, and NSGA-II is the dominant operator in the humanitarian logistics MOEA literature [11, 16], enabling direct comparison.

3.1 Chromosome Encoding Scheme

Each chromosome $\mathbf{C} = (\mathbf{X}, \mathbf{R}, \mathbf{A}, \mathbf{W})$ consists of four segments. The binary vector $\mathbf{X} \in \{0, 1\}^{|\mathcal{H}|}$ determines planned hub establishment. The continuous vector $\mathbf{R} \in [0, 1]^{|\mathcal{H}|}$ defines the inventory fill fraction, setting $q_k = R_k \cdot \kappa_k$. The integer vector $\mathbf{A} \in \{0, \ldots, |\mathcal{H}| - 1\}^{|\mathcal{I}|}$ encodes the preferred anchor hub for each demand i. The continuous vector $\mathbf{W} \in [0,1]^6$ contains six heuristic weights governing the decoder: w_0 and w_3 dictate demand sorting priority by urgency $(\lambda \cdot D)$ and isolation; w_1, w_2, w_4 score candidate hubs by travel time, residual inventory, and preference for planned over reactive hubs; w_5 controls conservatism towards opening new reactive hubs.

3.2 Priority-Based Decoding Heuristic

The decoder is invoked once per chromosome per scenario $s \in \mathcal{S}$, reconstructing all second-stage decisions from $\mathbf{C}$ in four steps.

Step 1 – Initialisation. Planned hubs with $X_k = 1$ and $r_{ks} \leq \chi$ are activated with pre-positioned inventory $q_k = R_k \cdot \kappa_k$. If none qualify, the safest hub is forced active.

Step 2 – Demand Prioritisation. Each demand node i receives a composite score prioritising urgency, isolation, and proximity to active hubs. Demands are then processed in descending score order.

Step 3 – Tiered Hub Assignment. Each demand i is assigned to the best reachable active hub, scored by travel time and residual capacity, starting from its anchor π_{A_i}. If all candidates are at capacity (*Pass 2*), the assignment still proceeds with a capacity violation. Only when no reachable active hub exists does the decoder open a new reactive hub as a last resort, incurring its setup cost and incrementing CV.

Step 4 – Supply Routing and Objective Accumulation. Origins supply the most under-supplied active hubs, and surplus is redistributed via discounted inter-hub trans-shipment (α). Expected objectives Z_1 and Z_2 are accumulated across all $s \in \mathcal{S}$. Algorithm 1 formalises these procedures.

With hub activations and demand assignments fixed by Steps 1–3, the supply routing in Step 4 reduces to a **minimum cost flow (MCF)** sub-problem: origins are sources, under-supplied hubs are sinks, and surplus inventory is redistributed via lateral transshipment arcs. Solving MCF exactly per scenario ensures that Z_1 is minimised to near-optimality for any decoded hub configuration, and is a key driver of PB-NSGA's solution quality.

3.3 Genetic Operators

The initial population seeds $\mathbf{X}$ with 1 to $\lfloor 0.6|\mathcal{H}| \rfloor$ open hubs, $\mathbf{R}$ from discrete pre-positioning levels, and $\mathbf{A}$, $\mathbf{W}$ uniformly. Crossover (probability p_c) applies Uniform Crossover to $\mathbf{X}$ and $\mathbf{A}$, and SBX (index η_{SBX}) to $\mathbf{R}$ and $\mathbf{W}$. Mutation (base rate p_m, decaying linearly from p_m^{high} to p_m^{low} over G generations) applies bit-flip, random-reset, and polynomial perturbation to the respective segments. Survival follows NSGA-II elitist constrained dominance [15]: feasibility first, then non-dominated rank, then crowding distance with Hamming tiebreaking in $\mathbf{X}$-space. Rank-1 stagnation triggers increased tournament size and $\mathbf{W}$-mutation, following the adaptive elitist strategy of [21].

Algorithm 1 Priority-Based Decoder (single scenario s)

Activate $\mathcal{H}_{\text{act}} \leftarrow \{k : X_k = 1,\ r_{ks} \leq \chi\}$; set $q_k \leftarrow R_k \kappa_k$
if $\mathcal{H}_{\text{act}} = \emptyset$ **then**
    Force-activate $\arg\min_k r_{ks}$ among $\{k : X_k = 1\}$
end if
Score demands by urgency, isolation, and proximity; sort descending
for each demand i (in priority order) **do**
    $K \leftarrow \max(1, \lceil w_5 |\mathcal{H}| \rceil)$; trial order $\leftarrow \pi_{A_i}$
    Pass 1: $k^* \leftarrow \arg\max$ HubScore(k) over first K active, reachable hubs with residual > 0
    if Pass 1 succeeds **then**
        Assign $i \rightarrow k^*$; deduct demand from residual inventory of k^*
    else if any active reachable hub remains **then**
        Pass 2: assign i to first such hub (capacity violation allowed)
    else
        Reactive fallback: open nearest safe inactive hub; add F^a_{ks}; $\mathrm{CV}^s \mathrel{+}= 1$
    end if
end for
Route supply from origins to hubs and redistribute surplus via MCF on the hub sub-network.
Accumulate Z_1^s, Z_2^s
return $(Z_1^s, Z_2^s, \mathrm{CV}^s)$

4 Computational Experiments

4.1 Experimental Setup

PB-NSGA is implemented in C++17; dataset construction and post-processing are performed in Python 3, on a workstation with an 11th Gen Intel Core i5-1135G7 @ 2.40 GHz and 16 GB RAM. Algorithm parameters are fixed for all runs: $N = 200$, $G = 300$ (CV-Small) / 500 (CV-Large), $p_c = 0.98$, $p_m^{\text{high}} = 0.40$, $p_m^{\text{low}} = 0.10$ (linear decay), $\eta_{\text{SBX}} = 1.5$, $\eta_{\text{poly}} = 8$. The parameters are chosen via a grid search process. Problem constants are $\chi = 0.7$, $\alpha = 0.6$, $\gamma = 3.0$ kg/person, $\lambda_0 = 0.8$. Each configuration is executed for 20 independent runs, reporting mean $\pm$ std. We use **Hypervolume (HV)** [22] and **IGD$^+$** [23] as performance metrics, computed against the combined non-dominated reference front pooled across all algorithms.

4.2 Dataset Description

Central Vietnam Dataset. We construct two instances modelling four flood-prone provinces: Da Nang, Quang Nam, Thua Thien-Hue, and Quang Ngai. **CV-Small** uses district-level demand nodes ($|\mathcal{I}| = 20$, $|\mathcal{H}| = 5$, $|\mathcal{J}| = 2$, $|\mathcal{S}| = 3$); **CV-Large** uses commune-level resolution ($|\mathcal{I}| = 100$, $|\mathcal{H}| = 20$, $|\mathcal{J}| = 12$, $|\mathcal{S}| = 3$). Node coordinates map to real administrative centroids, confirmed logistics facilities, and vital supply origins. Scenario risks and road disruptions are calibrated using a multi-criteria flood vulnerability score [24]. Three transport modes – trucks, motorboats, and helicopters – are considered, with helicopter links capped at 15% of active routing connections per scenario.

Scenario generation via SAA [25]. To improve coverage of plausible flood conditions, scenarios are drawn from a profile bank spanning mild, severe, and extreme flood regimes with varying epicenter count, intensity, road disruption rate, and circuity coefficient. The CV-Large case study further validates the first-stage decisions $(\hat{x}, \hat{q})$ on an independent out-of-sample (OOS) set of $N' = 10$ adversarial scenarios, disjoint from training, to assess generalization under catastrophic conditions. Because this OOS holdout is deliberately extreme-only, we treat feasibility as the primary robustness metric and interpret large OOS increases in Z_2 as stress amplification under adversarial conditions rather than ordinary prediction error.

Scenario-count sensitivity. Figure 1 reports a scenario-count sensitivity study ($K = 10$ replications, exact enumeration on CV-Small). The std of Z_2 contracts from ±110k ($N = 3$) to ±30k ($N = 30$) and HV std from 0.25 to 0.07, confirming SAA variance reduction; a transient peak at $N = 5$ reflects unlucky extreme-heavy draws. Our calibrated three-profile design achieves equivalent stability at one-third the scenario count.

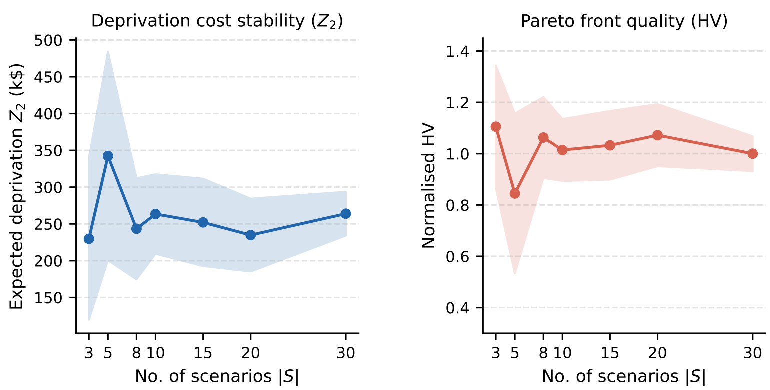

Fig. 1. Scenario-count sensitivity on CV-Small ($K = 10$ replications, exact enumeration). Std of Z_2 and HV decrease with N, confirming variance contraction; our calibrated three-profile design achieves stability equivalent to $N \approx 10$ random draws.

4.3 Experiment 1: Baseline Comparison

We evaluate PB-NSGA against four baselines on the **CV-Small** instance. The **Greedy Heuristic** simulates ad-hoc decision-making: it constructs a council of managers with different cost-safety weight profiles, each greedily selecting hubs and assigning demands by nearest-neighbour logic, then returns their combined non-dominated solutions. **VNS-TS** [10] is a Variable Neighbourhood Search with Tabu Search, adapted from the flood relief literature as a strong meta-heuristic baseline. **GWO-HD** [11] is a Grey Wolf Optimizer with Hamming Distance moves, representing a recent meta-heuristic designed for discrete hub-location problems. The **MILP** solver [26] applies the Adaptive Weighted Sum method [27] to systematically sweep the objective space and produces the reference Pareto front for CV-Small.

Table 1. Baseline comparison on CV-Small. HV and IGD$^+$ are normalized w.r.t. the combined non-dominated reference front across all algorithms. Best values in **bold**.

Algorithm	HV (norm.) ↑	IGD$^+$ ↓	CPU Time (s) ↓
Greedy Heuristic	0.292	0.333	$<$ **0.1**
VNS-TS	0.777	0.117	60.0
GWO-HD	0.780	0.152	0.6
MILP (Adaptive WS)	**0.997**	**0.005**	4.9
PB-NSGA (Ours)	0.925	0.039	1.1

Table 1 reveals a clear performance hierarchy. The MILP solver defines the reference Pareto front for CV-Small, establishing the quality ceiling at $\mathrm{HV} = 0.997$. The Greedy heuristic runs near-instantly but remains far from the front. VNS-TS and GWO-HD both achieve competitive HV ($\approx$0.78) with VNS-TS requiring 60 s and GWO-HD only 0.6 s. PB-NSGA closes the gap to MILP substantially ($\mathrm{HV} = 0.925$, $\mathrm{IGD}^+ = 0.039$) in 1.1 s, demonstrating that the priority-based decoder effectively navigates the combinatorial structure of MO-IHLNDP.

4.4 Experiment 2: Case Study on Central Vietnam

We apply PB-NSGA, GWO-HD, and VNS-TS to the full-scale **CV-Large** instance ($|\mathcal{I}| = 100$, $|\mathcal{H}| = 20$), where exact solvers become intractable. All other parameters follow Sec. 4.1.

Figure 2 shows Pareto fronts for both instances. On CV-Small, PB-NSGA closely tracks the MILP reference; GWO-HD converges to a similar region with more scatter; VNS-TS covers only the cost-efficient tail. On CV-Large (power scale), PB-NSGA dominates the low-cost, low-deprivation region (Z_1: 4.3M–5M, Z_2: 41K–45K); GWO-HD extends the front rightward at the cost of higher Z_1 (up to 26M); VNS-TS is confined to the high-cost tail ($Z_1 = 81$M, $Z_2 = 150$K).

On CV-Large across 20 matched runs per algorithm, PB-NSGA achieves $\mathrm{HV} = 1.000 \pm 0.006$ and $\mathrm{IGD}^+ = 0.000 \pm 0.001$ in 8.7 s on average. GWO-HD reaches $\mathrm{HV} = 0.982 \pm 0.007$ in 4.8 s, while VNS-TS improves to $\mathrm{HV} = 0.823 \pm 0.038$ but still requires 180.6 s per run – a 20.7× runtime disadvantage relative to PB-NSGA. PB-NSGA therefore combines the best frontier quality with sub-10-second runtime on the full-scale case study.

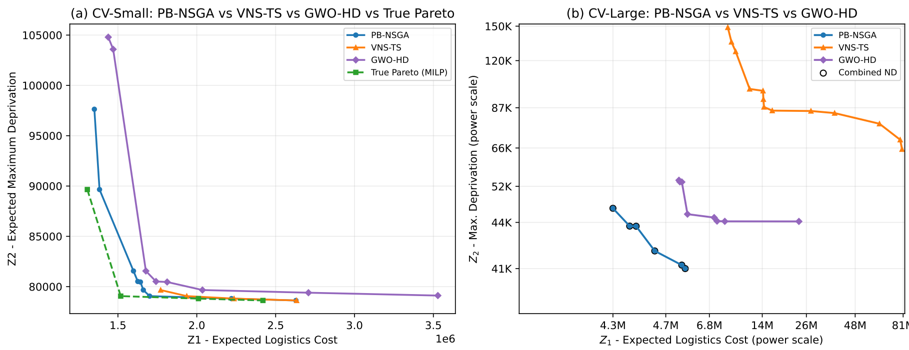

Fig. 2. Combined Pareto fronts for PB-NSGA, GWO-HD, and VNS-TS. **Left (CV-Small):** all three algorithms compared against the MILP true Pareto front. **Right (CV-Large, power scale):** combined non-dominated front across 20 matched runs for each algorithm. Z_1: expected logistics cost; Z_2: expected maximum deprivation cost.

Because all three algorithms share the same decoder, the 100% SAA-100 feasibility rate across all 20 seeds is attributable to the recourse engine, not the search. On the adversarial OOS set (10 extreme scenarios), PB-NSGA alone maintains perfect feasibility across all seeds; GWO-HD and VNS-TS remain highly feasible at 96.4% (95% CI [93.4%, 98.8%]) and 97.8% (95% CI [96.2%, 99.2%]), respectively. The feasibility gap reflects search quality, not recourse privilege.

Figure 3 visualizes the knee-point solution (Z_1 = \$4.6M, Z_2 = 43,453). The eight planned hubs remain active with identical fill levels across all three scenarios; no reactive hub is activated. This is an optimal structural property of dense networks: pre-positioning at eight commune-resolution locations mathematically dominates reactive setup costs (\$51K–\$109K per scenario), while the recourse mechanism (Algorithm 1, Pass 3) remains available and verifiably triggers on sparser CV-Small instances. Adaptation across severity regimes is modal: helicopter routes displace truck links in the extreme scenario.

H16 (91%) and H15 (89%) carry the heaviest pre-positioned loads, anchoring coverage for the central and southern coastal corridor. The primary adaptation across scenarios is modal: mild and severe conditions are served by a mix of truck and watercraft links, while the extreme scenario activates helicopter routes (orange dashed) to reach high-risk demand nodes that become inaccessible by road and water.

As a sensitivity check on training-set size, we compare the calibrated three-scenario design against a tier-balanced $N = 10$ training set using 20 PB-NSGA runs each. The calibrated $N = 3$ design achieves significantly better OOS feasibility (1.000 vs. 0.931; $\Delta = -0.069$, 95% CI $[-0.128, -0.029]$) at one-third the training burden. By contrast, the mean OOS ΔZ_2 difference is not significant ($\Delta = -437\%$, 95% CI $[-1865\%, +1969\%]$) because the $N = 10$ design is highly right-skewed: even after excluding five BIG_M sentinel evaluations, one

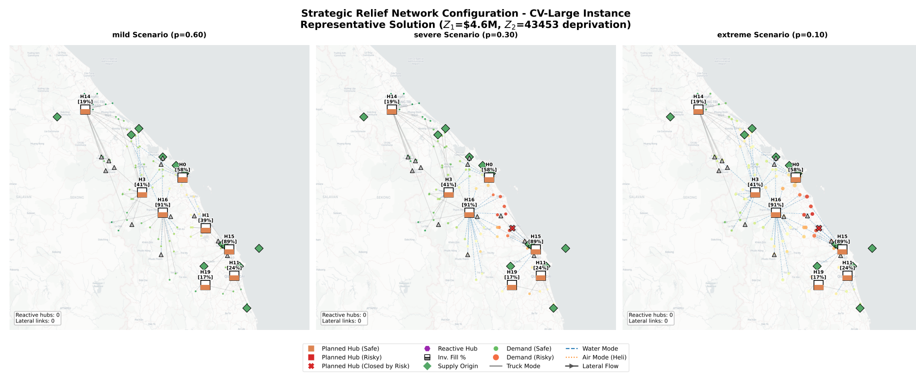

Fig. 3. Strategic relief network for a representative solution on CV-Large ($Z_1 = \$4.6$M, $Z_2 = 43{,}453$) across three probabilistic scenarios: mild ($p = 0.60$), severe ($p = 0.30$), and extreme ($p = 0.10$). Hub inventory fill levels are annotated in brackets. No reactive hub is activated in any scenario. Line styles indicate transport mode: solid gray (truck), blue dashed (water), orange dashed (helicopter).

seed still exhibits a genuine catastrophic OOS failure (22,037% mean OOS ΔZ_2). We therefore treat feasibility as the primary robustness conclusion and interpret the $N = 10$ result as evidence that uniform-probability random training carries substantially higher robustness variance under adversarial extreme-only stress.

Decision support. Hub selection frequency across all 20 CV-Large runs (128 Pareto solutions) reveals a clear investment hierarchy: Quang Ngai Port (H15) and Phuoc Son Helipad (H16) appear in every solution (100%), followed by Huong Viet Depot (H14, 82.8%), Da Nang Airport (H0, 77.3%), Tam Ky Logistics Hub (H1, 67.2%), Binh Son Warehouse (H9, 59.4%), and Dong Giang Station (H3, 57.8%). H1's risk surges from 0.46 (mild) to 0.94 (severe), exceeding $\chi = 0.7$ and forcing deactivation under high-severity scenarios; authorities should treat it as a conditional asset and pre-stage contingency inventory at H14 (risk $\leq$ 0.13 across all scenarios). Under the extreme scenario, road disruption enforces a modal shift to watercraft and helicopter routes; pre-equipping H15 and H16 with marine terminals and maintaining rotary-wing access at H0 delivers asymmetric resilience at lower cost than redundant infrastructure.

5 Conclusion and Future Work

This paper makes three contributions to humanitarian logistics under uncertainty. First, we formulate MO-IHLNDP, a two-stage stochastic bi-objective model that jointly captures incomplete hub networks, multi-modal transport, deprivation cost, and pre-positioned inventory for flood disaster relief. Second, we propose PB-NSGA, a priority-based NSGA-II with an MCF-backed decoder that efficiently navigates the coupled first- and second-stage decisions. Because the same decoder and evaluation core is also embedded in GWO-HD and VNS-

TS, the matched 20-seed CV-Large study isolates search from reconstruction: all three methods remain highly feasible under adversarial OOS stress, but PB-NSGA alone maintains 100% OOS feasibility while delivering the best Pareto quality in under 10 s. Third, applied to four flood-prone provinces in Central Vietnam, the joint framework delivers a practical decision-support tool: a quantified cost-deprivation trade-off, a stable hub investment hierarchy anchored by the Quang Ngai Port Hub and Phuoc Son Helipad, explicit identification of resilience gaps for contingency planning, and evidence that the calibrated three-scenario design is more reliable than a tier-balanced $N = 10$ alternative at one-third the training burden.

Limitations and future work. The model is validated on a synthetic instance calibrated from administrative and geographic data; field validation against actual disaster records is necessary before operational deployment. Three directions are prioritised: integrating real-time sensor and satellite feeds into a live decision-support ecosystem; extending the model to explicitly penalize hub processing degradation under high-risk conditions, bridging scenario-based planning and tail-risk management; and scaling PB-NSGA to hour-long runtime budgets with continuous convergence guarantees for large-scale regional deployments.

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